

Quick Check

Lesson 4: Tangent and the Sine–Cosine–Tangent Relationship: Solving the Mango-Tree Problem

10 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.1: Trigonometry I | Lesson 4

In Kisumu, a student measures the sun's angle with the ground as 53° . The mango tree's shadow on the ground is 9.0 m long. She wants the tree's height. The tree is vertical and the ground is horizontal. Which ratio links the two lengths she knows or wants (height and shadow) with the 53° angle?

- A** No ratio works unless she first measures the length of the sun's ray (the hypotenuse)
- B** $\sin 53^\circ$, because sine is the most useful ratio for every triangle
- C** $\cos 53^\circ$, because the shadow is on the ground
- D** $\tan 53^\circ$, because it links the opposite side (height) and the adjacent side (shadow)

In a right-angled triangle, $\sin \theta = O/H$ and $\cos \theta = A/H$. What do we get when we simplify $(O/H) \div (A/H) = (O/H) \times (H/A)$?

A O/A , which is $\tan \theta$

B O/H

C A/O

D H/A

A right-angled triangle has sides of 6 cm, 8 cm and 10 cm (the hypotenuse). For the angle θ opposite the 6 cm side, what is $\tan \theta$?

A 1.33

B 0.75

C 0.6

D 0.8

From mathematical tables, $\sin 40^\circ = 0.6428$ and $\cos 40^\circ = 0.7660$. Using $\tan \theta = \sin \theta \div \cos \theta$, what is $\tan 40^\circ$ to 3 decimal places?

A 1.409

B 0.839

C 1.192

D 0.492

A pair finds that for one angle, $\sin \theta \div \cos \theta$ is 0.8391 and $\tan \theta$ read from tables is 0.8390. What is the most sensible explanation of this tiny difference?

- A They used the wrong triangle, because $\tan$ changes with the size of the triangle
- B The identity only works for 45°
- C Table values are rounded, so tiny differences are expected
- D The identity $\tan \theta = \sin \theta / \cos \theta$ does not work for that angle

A learner says, "We can already get $\tan \theta$ from $\sin \theta$ and $\cos \theta$, so $\tan$ is unnecessary." What is the best reply?

- A** Tan can only be used when the hypotenuse is known
- B** Tan is only used for angles bigger than 90°
- C** They are right: $\tan$ is redundant and is never needed
- D** Tan is a convenient shortcut when we know an angle and one side but not the hypotenuse

The sun makes an angle of 53° with the ground. A vertical mango tree of height h casts a 9.0 m shadow along the ground, and the 53° angle is at the end of the shadow, between the ground and the sun's ray. Which statement is correct?

- A** The 9.0 m shadow is the side opposite the 53° angle, and the tree height h is the side adjacent to it
- B** Both the shadow and the tree are adjacent to the 53° angle
- C** The 9.0 m shadow is the side adjacent to the 53° angle, and the tree height h is the side opposite it
- D** The 9.0 m shadow is the hypotenuse

A mango tree has a shadow 9.0 m long when the sun makes an angle of 53° with the ground. Given $\tan 53^\circ = 1.3270$ and $\tan 53^\circ = h \div 9.0$, what is the height h of the tree?

A 5.42 m

B 11.943 m

C 7.19 m

D 6.78 m

A student is 1.6 m tall and her shadow is 1.2 m long. At the same time, a mango tree has a shadow 9.0 m long. Using similar triangles, $(\text{tree height}) \div 9.0 = 1.6 \div 1.2$. What is the height of the tree?

A 14.4 m

B 12.0 m

C 9.4 m

D 6.75 m

On the 'Ratio Web' in the Trigonometry Model, $\tan \theta$ is linked to $\sin \theta$ and $\cos \theta$. For an angle of 53° , $\sin 53^\circ \approx 0.799$ and $\cos 53^\circ \approx 0.602$. Using the identity, what is $\tan 53^\circ$ to 3 decimal places?

A 1.327

B 0.481

C 0.753

D 1.401

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).